

**Analysis of Terephthalic Acid Process Samples, PTA-Slurry and PTA-Condensate, By
APCI/LC/MS and GC/MS (atmospheric pressure chemical ionization; liquid chromatography-
mass spectrometry; gas chromatography-mass spectrometry)**

SUMMARY

Two samples from the purified terephthalic acid (PTA) process step in the INCA Ottana plant were submitted for identification of major and minor components by mass spectrometry. The samples were labeled PTA-slurry, D2403-6/3/99 and PTA-condensate, E2403 6/28/99. The PTA-slurry sample was separated into a mother liquor liquid phase and solid phase. These separate phases and the condensate sample were analyzed by atmospheric pressure chemical ionization liquid chromatography-mass spectrometry (APCI/LC/MS) in the negative ion (NI) mode and by atmospheric pressure chemical ionization liquid chromatography-mass spectrometry-mass spectrometry (APCI/LC/MS/MS). The separate phases and the condensate sample were also derivatized with N,O-bis-(tri-methyl-silyl)-tri-fluoro-acetamide (BSTFA) and analyzed by gas chromatography-mass spectrometry (GC/MS) in the electron impact (EI) and positive ion ammonia chemical ionization (PCI-NH₃) modes. The mother liquor phase of the PTA-slurry sample and the PTA-condensate sample were extracted with methylene chloride and analyzed by GC/MS in the EI mode. Thirty, five, and ten components were detected and tentatively identified in the PTA-mother liquor, PTA-solid, and, PTA-condensate, respectively.

EXPERIMENTAL

1 - APCI/LC/MS

Fifty- μ l aliquots of the PTA-mother liquor sample neat, fifty- μ l aliquots of the PTA mother liquor condensate sample neat and fifty μ l aliquots of the PTA solid saturated in water/acetonitrile were injected on a Waters MS 600 ternary gradient liquid chromatograph coupled to a Finnigan TSQ700, SN TS000157, quadrupole MS/MS system via a Finnigan atmospheric pressure chemical ionization (APCI) interface operating in the NI mode. The following analysis conditions were used:

Column:	250 X 4.6 mm ID 5 μ m Zorbax Rx-C18
Mobile phase:	95/5 A/B to 75/25 A/B at 20 min. to 70/30 A/B at 30 min. to 100B at 38 min. A = 0.5 v% acetic acid in water B = 0.5 v% acetic acid in acetonitrile
Flow:	1.0 ml/min
UV detection:	254 nm



APCI conditions:	Capillary- 160°C	Corona- 3.6 kV; 5µA
	Sheath- 80 psi N ₂	Aux -20
MS conditions:	APICID - +10V	Vaporizer -400°C
	Electron multiplier- 1400V	Pre-amp gain- 10E ⁻⁸ A/V
	Conversion Dynode - +15 kV	Mode - -Q3MS; LMR; cent
	Scan- 120 to 1000 amu (-)	Rate- 1.0 s/scan

2 - APCI/LC/MS/MS

Fifty-µl aliquots of the PTA-mother liquor and PTA condensate were re-injected neat into the system but this time in the APCI/LC/MS/MS/NI mode. The above analysis conditions were used with the following changes and/or additions:

Electron multiplier- 2800 V	Pre-amp gain- 10E ⁻⁸ A/V
Collision gas - 2.0 mtorr Argon	Collision energy - 30 eV

Lists of Daughter Ion scans are filed with this report

3 - GC/MS (derivatization)

One-ml aliquots of PTA-mother liquor and the PTA condensate were evaporated to dryness under nitrogen. These residues and a fifty-microgram portion of the PTA-solid were dissolved in one-milliliter of pyridine and reacted with one milliliter of BSTFA. One-µl aliquots of the solutions were injected onto a Finnigan SSQ7000, SN TS010023, quadrupole GC/MS system in EI and PCI-NH₃ modes. The following analysis conditions were used:

Column:	30m x .32 mm (0.25 µm film) Zebron ZB-5
Temperatures:	Column 110°C (2 min) to 310°C at 8°/minute
	Injector - 300°C Interface - 300°C Ionizer - 180°C
Flow:	Linear velocity - 40 cm/s Split - 20:1
Detector:	Electron multiplier - 1300 / 1400 V (EI/CI)
	Conv. Dyn. - (-)15 kV Mode - +Q1MS, CENT
	Electron Energy - 70eV (EI) 200eV (CI)
	Scan - 35 to 650 amu (+EI) 60 to 650 amu (+CI)
	Rate - 1.0 s/scan
Source Pressure:	1600 mtorr ammonia



4 - GC/MS (extraction)

Three-ml aliquots of PTA-mother liquor and the PTA condensate were extracted with one-ml aliquots of methylene chloride. One- μ l aliquots of the extracts were injected onto a Finnigan SSQ7000, SN TS010023, quadrupole GC/MS system in EI mode. The above analysis conditions were used with the following changes:

Temperatures: Column 50°C to 310°C at 16°/minute

RESULTS

A chromatogram with UV detection and computer reconstructed total ion chromatogram from the APCI/LC/MS/NI analysis of PTA-Mother Liquor, D2403-6/3/99, are presented in Figure 1. A summary of mass spectral data plus tentative assignments from APCI/LC/MS and APCI/LC/MS/MS analyses is presented in Table I. A total ion chromatogram computer reconstructed from the GC/MS/EI analysis of PTA-Mother Liquor, D2403-6/3/99, reacted with BSTFA is presented in Figure 2. A summary of mass spectral data including tentative assignments from the GC/MS/EI and CI analyses is presented in Table II. A computer reconstructed total ion chromatogram from the GC/MS/EI analysis of PTA-Mother Liquor, D2403-6/3/99, extracted with methylene chloride is presented in Figure 3. A summary of mass spectral data including tentative assignments from the GC/MS/EI analysis is presented in Table III.

A chromatogram with UV detection and computer reconstructed total ion chromatogram from the APCI/LC/MS/NI analysis of PTA-solid, D2403-6/3/99, are presented in Figure 4. A summary of mass spectral data plus tentative assignments from APCI/LC/MS analysis is presented in Table IV. A computer reconstructed total ion chromatogram from the GC/MS/EI analysis of PTA-solid, D2403-6/3/99, reacted with BSTFA is presented in Figure 5. A summary of mass spectral data including tentative assignments from the GC/MS/EI and CI analyses is presented in Table V.



A chromatogram with UV detection and computer reconstructed total ion chromatogram from the APCI/LC/MS/NI analysis of PTA-condensate, E2403 6/28/99, are presented in Figure 6. A summary of mass spectral data plus tentative assignments from APCI/LC/MS and APCI/LC/MS/MS analyses is presented in Table VI. A computer reconstructed total ion chromatogram from the GC/MS/EI analysis of PTA-condensate, E2403 6/28/99, reacted with BSTFA is presented in Figure 7. A summary of mass spectral data including tentative assignments from the GC/MS/EI and CI analyses is presented in Table VII. A computer reconstructed total ion chromatogram from the GC/MS/EI analysis of PTA-condensate, E2403 6/28/99, extracted with methylene chloride is presented in Figure 8. A summary of mass spectral data including tentative assignments from the GC/MS/EI analysis is presented in Table VIII. Raw data and mass spectra are filed with this report.

REFERENCES

- B. M. Bell Finn** *SSQ7000 Logbook 199901070-64 and 68.*
- B. M. Bell Finn** *TSQ700 Logbook 1999-10 and 12.*
- B. M. Bell** *Raw Data Envelope* GL-AL MD-1999-010109.



FIGURE 1

A chromatogram with UV detection plus a computer reconstructed total ion chromatogram from the APCI/LC/MS/NI analysis of PTA-Mother Liquor, D2403-6/3/99.

CHRO:	mt7c10109bb2	29-JUN-99	Elapse:	12:13.3	710
Samp:	PTA slurry Mother Liq D2403 6/3/99		Start:	10:15:36	2619
Comm:	-APCI/LC/MS 250x4.6 Zorbax Rx-C18 5u H2O/ACN/AA grad@1mL/min				
Mode:	APCI-Q3MS APICID LMR UP LR		Study:	MD-1999-010109	
Oper:	BB/TSQ700	Client: Piras	Inlet:	LC	
Peak:	1000.00 mmu	Label wndw: 1 > 2620	Masses:	120 > 1000	
Area:	0, 4.00, 0	Baseline: 0, 3	Label:	0, 40.0	

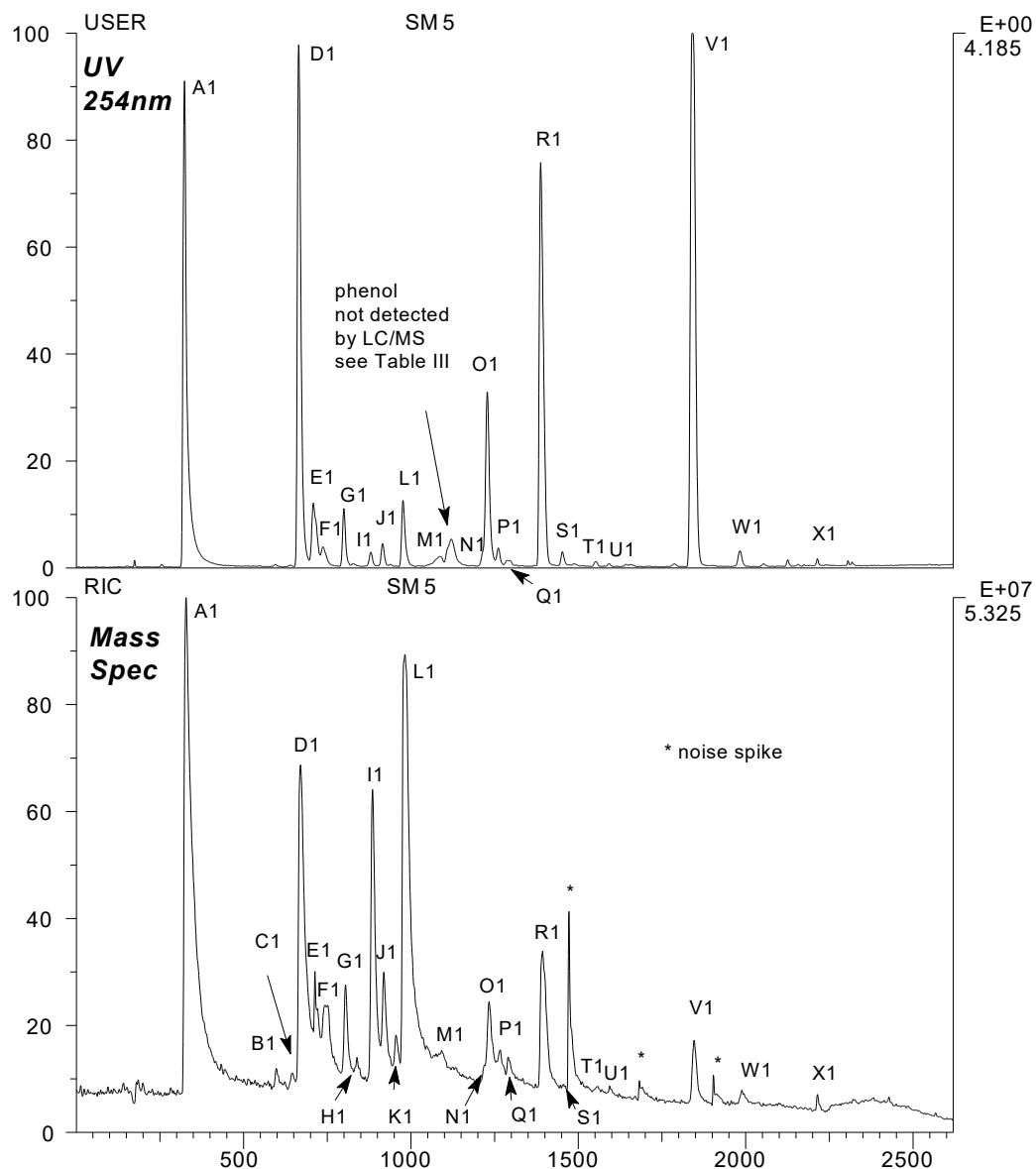


TABLE I

Summary of mass spectral data from the APCI/LC/MS and APCI/LC/MS/MS of PTA-Mother Liquor, D2403-6/3/99.

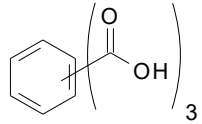
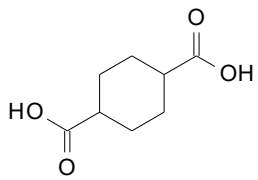
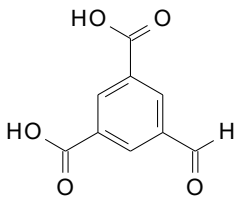
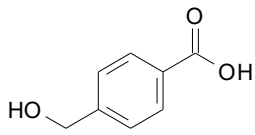
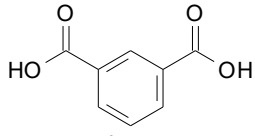
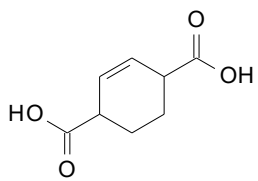
Peak ID	m/z Observed APCI/NI	MS/MS of m/z	m/z Observed daughters	MW	Tentative Assignment
A1	209,191,383 (M-H) ⁻ , (M-H-H ₂ O) ⁻ , (2M-H) ⁻ , Benzoic acid	209	121,77,165,93,209	210	 or isomer
B1	231,171,291 (M+CH ₃ CO ₂) ⁻ , (M-H) ⁻ , Cyclohexane dicarboxylic acid	231	127,153,84,78,171	172	 or isomer
C1	193,253 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ , Trimesic acid	193	77,121,149,165,105	194	 or isomer
D1	211,151 (M+CH ₃ CO ₂) ⁻ , (M-H) ⁻ , 4-hydroxybenzoic acid	151	107,89,151,121,77	152	 or isomer
E1	151 (M-H) ⁻ ,	151	107,89,151,121,77	152	isomer of D1
F1	165,229,169,331,289 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ , (M-H) ⁻ , Phthalic acid	165 229	77,121,165 125,169,149	166 170	 or isomer +  or isomer

TABLE I

**Summary of mass spectral data from the APCI/LC/MS and APCI/LC/MS/MS of
PTA-Mother Liquor, D2403-6/3/99.**

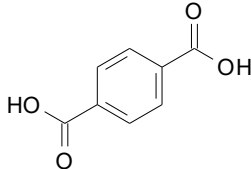
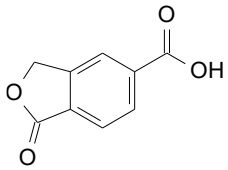
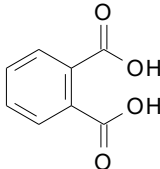
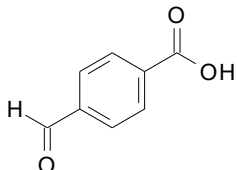
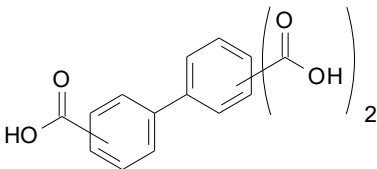
Peak ID	m/z Observed APCI/NI	MS/MS of m/z	m/z Observed daughters	MW	Tentative Assignment
G1	165,225 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ , Terephthalic acid	165	121,77,165,93	166	 or isomer
H1	149,209 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	149	77,	150	isomer of M1
I1	177,237 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ , 5 carboxybenzofuranone	177	133,89,177	178	 or isomer
J1	177,237,355 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	177	133,89,177	178	isomer of I1
K1	171,231 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	171	127,171	172	isomer of B1
L1	165,225,331 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	165	121,165,77,93	166	 or isomer
M1	149,209 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	149	105,121,149, 89	150	 or isomer
N1	253,193 (M+CH ₃ CO ₂) ⁻ , (M-H) ⁻ ,	193	149,121,105	194	isomer of C1
O1	285,345, (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ , Biphenyl-4,4'-dicarboxylic acid	285	197,153,241, 181,285,213, 121	286	 or isomer



TABLE I

**Summary of mass spectral data from the APCI/LC/MS and APCI/LC/MS/MS of
PTA-Mother Liquor, D2403-6/3/99.**

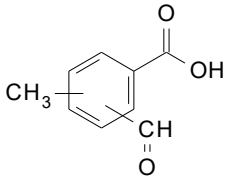
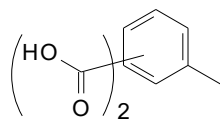
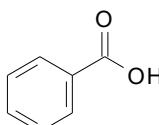
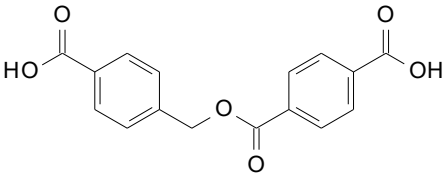
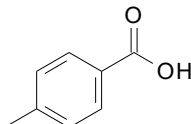
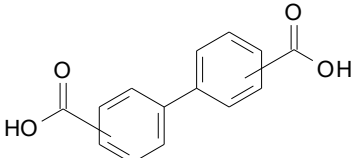
Peak ID	m/z Observed APCI/NI	MS/MS of m/z	m/z Observed daughters	MW	Tentative Assignment
P1	163,223 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	163	119,135	164	 or isomer
Q1	179,239 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	239	135,179,131	180	 or isomer
R1	181,121 (M+CH ₃ CO ₂) ⁻ , (M-H) ⁻ ,	121	77,121,93,65	122	 or isomer
S1	299,359 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ , 4,4'-dicarboxybenzylbenzoate	ND		300	 or isomer
T1	241,301 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	241	153,197,137	242	isomer of W1
U1	199 (M-H) ⁻ ,	199	171,199,117	200	No ID
V1	135,195 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	135	91,135	136	 or isomer
W1	241,301 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ , Biphenyl-dicarboxylic acid	241	153,197,169, 241	242	 or isomer
X1	185 (M-H) ⁻ ,	185	167,185	186	No ID

FIGURE 2

**A computer reconstructed total ion chromatogram from the GC/MS/EI analysis of
PTA-Mother Liquor, D2403-6/3/99, reacted with BSTFA.**

CHRO:	ms7k10109bb3	28-JUN-99	Elapse:	28:14	1689
Samp:	PTA-Moth. Liq. D2403 (BSTFA/pyridine)		Start:	09:16:37	2007
Comm:	EI/GC/MS 30mx0.25mmx0.25u ZB-5 110-8-310 m/z 35-650@1 20/1				
Mode:	EI+Q1MS LMR UP LR		Study:	MD-1999-010109	
Oper:	BB/SSQ7000	Client: Piras	Inlet:	GC	
Peak:	1000.00 mmu	Label wndw: 1 > 1500	Masses:	35 > 650	
Area:	0, 4.00, 0	Baseline : 0, 3	Label :	0, 40.0	

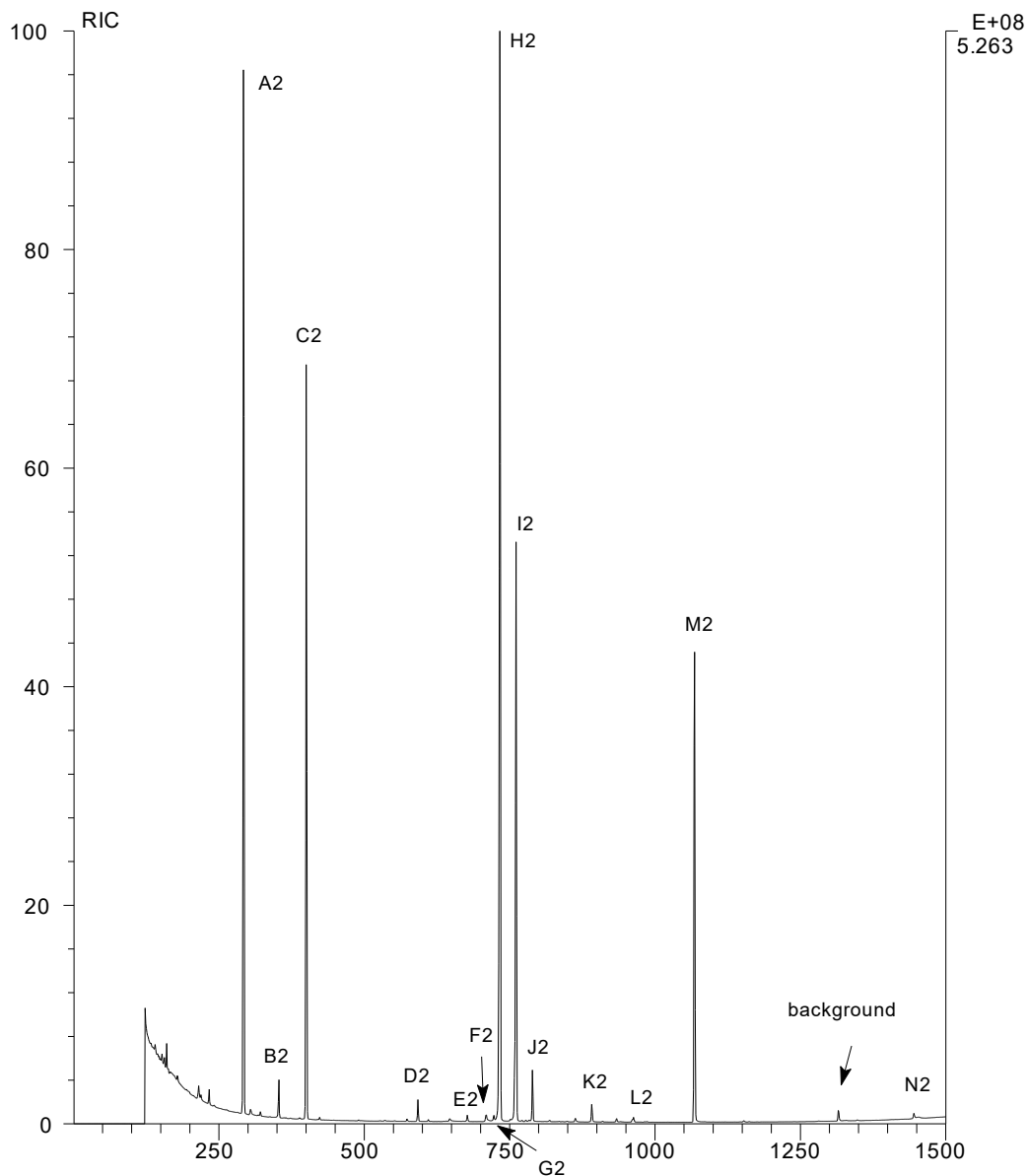


TABLE II

**Summary of mass spectral data from the GC/MS/EI and PCI-NH₃ analyses of
PTA-Mother Liquor, D2403-6/3/99, reacted with BSTFA.**

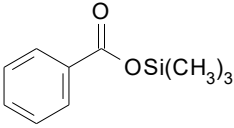
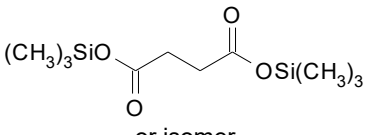
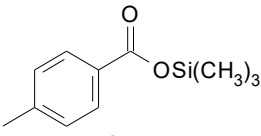
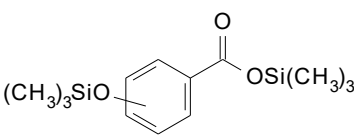
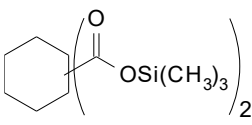
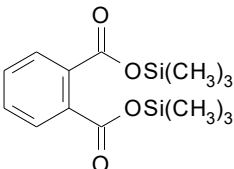
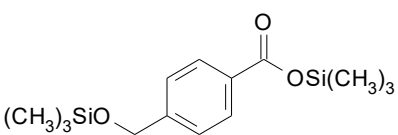
Peak ID	m/z Observed EI	m/z Observed PCI-NH ₃	MW	Tentative Assignment
A2	179,135,105,77,194,51	212,195,179,90 (M+NH ₄) ⁺ , (M+H) ⁺	194	 or isomer
B2	147,247,73,75,129,172,218,116,55,132	263,280 (M+H) ⁺ , (M+NH ₄) ⁺	262	 or isomer
C2	193,149,119,91,208,65	226,209,90,193 (M+NH ₄) ⁺ , (M+H) ⁺	208	 or isomer
D2	73,267,282,223,91,45,193	300,283 (M+NH ₄) ⁺ , (M+H) ⁺	282	 or isomer
E2	73,75,81,147,45,109,301	334,317,73 (M+NH ₄) ⁺ , (M+H) ⁺	316	 or isomer
F2	147,295,73,310,221,140,265,220	311,295 (M+H) ⁺	310	 or isomer
G2	73,75,81,147,45,109,301	334,317,73 (M+NH ₄) ⁺ , (M+H) ⁺	316	isomer of E2
H2	207,281,206,73,147,179,296,133,89,163,237,251	314,297 (M+NH ₄) ⁺ , (M+H) ⁺	296	 or isomer



TABLE II

**Summary of mass spectral data from the GC/MS/EI and PCI-NH₃ analyses of
PTA-Mother Liquor, D2403-6/3/99, reacted with BSTFA.**

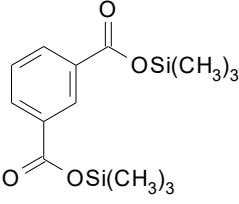
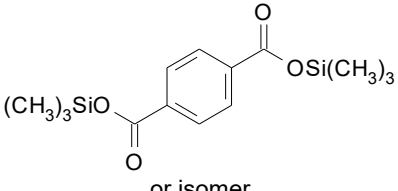
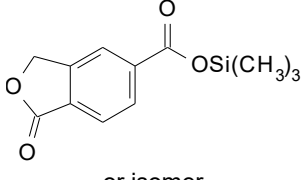
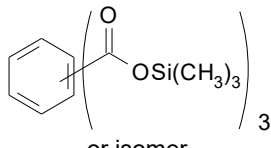
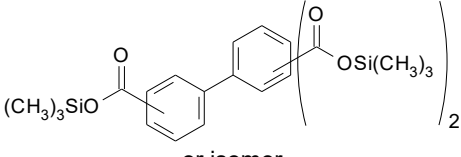
Peak ID	m/z Observed EI	m/z Observed PCI-NH ₃	MW	Tentative Assignment
I2	295,279,73,310,221,205,207,103,140,177	328,311,295 (M+NH ₄) ⁺ , (M+H) ⁺ ,	310	 or isomer
J2	295,221,103,251,140,73,135,178,310	328,311,295 (M+NH ₄) ⁺ , (M+H) ⁺ ,	310	 or isomer
K2	235,89,75,161,191,103,133,63,43	268 (M+NH ₄) ⁺	250	 or isomer
L2	221,89,63,76,51,179,165,236	254 (M+NH ₄) ⁺	236	No ID
M2	147,411,73,337,309,307,426,249,133	427,411,337 (M+H) ⁺ ,	426	 or isomer
N2	487,73,502,236,179,207,297,281,413,384,151	520,503 (M+NH ₄) ⁺ , (M+H) ⁺ ,	502	 or isomer



FIGURE 3

**A computer reconstructed total ion chromatogram from the GC/MS/EI analysis of
PTA-Mother Liquor, D2403-6/3/99, extracted with methylene chloride.**

CHRO:	ms7k10109bb1	28-JUN-99	Elapse:	02:48.6	164
Samp:	PTA-Moth. Liq. D2403 (3:1 CH ₂ Cl ₂ extr)		Start:	06:36:21	1531
Comm:	EI/GC/MS 30mx0.25mmx0.25u ZB-5 50-16-300 m/z 35-650@1 20/1				
Mode:	EI+Q1MS LMR UP LR		Study:	MD-1999-010109	
Oper:	BB/SSQ7000	Client: Piras	Inlet:	GC	
Peak:	1000.00 mmu	Label wndw: 1 > 1000	Masses:	35 > 650	
Area:	0, 4.00, 0	Baseline : 0, 3	Label :	0, 40.0	

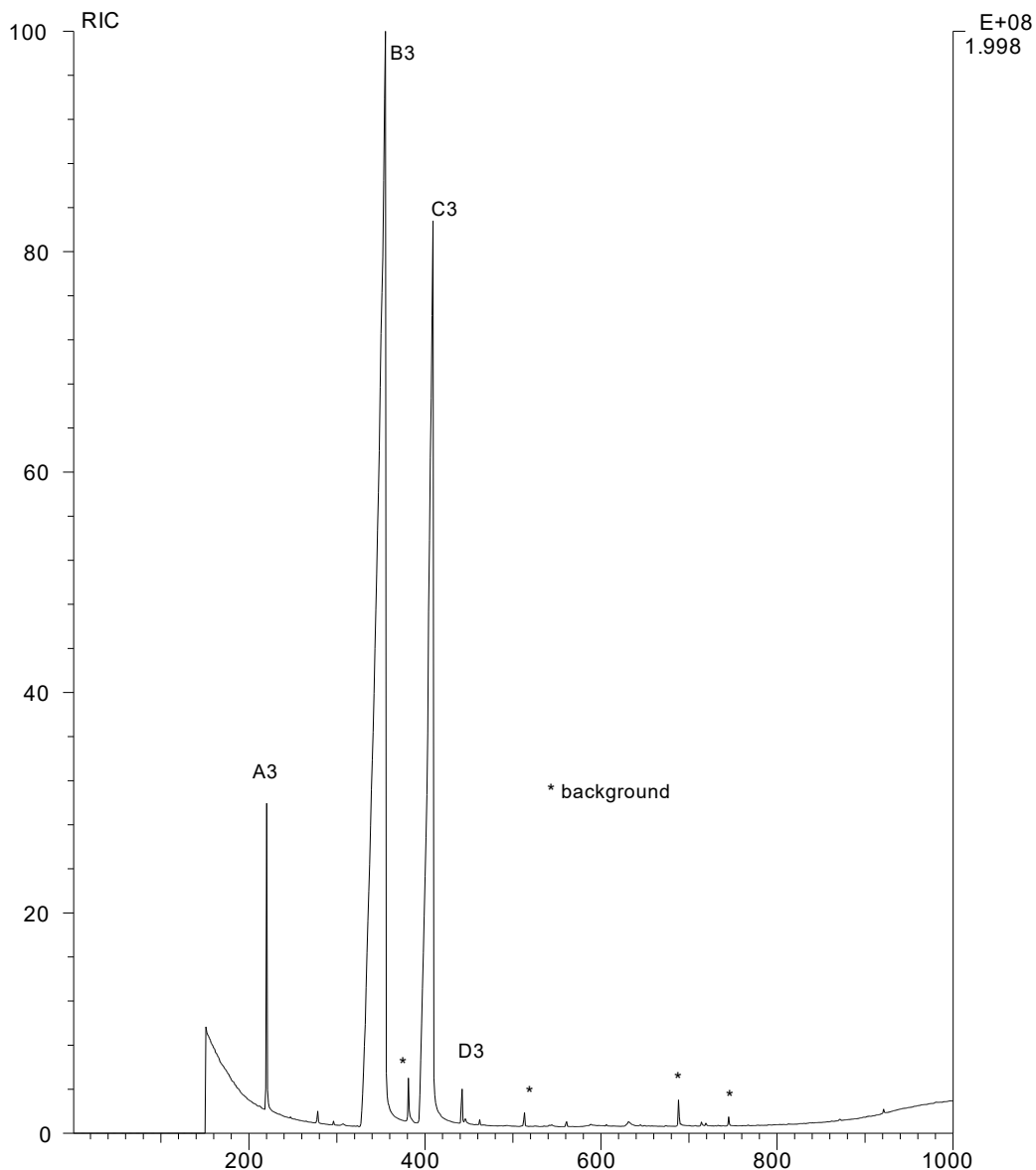


TABLE III

**Summary of mass spectral data from the GC/MS/EI analysis of
PTA-Mother Liquor, D2403-6/3/99 extracted with methylene chloride.**

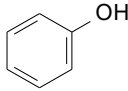
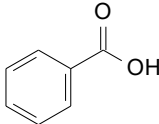
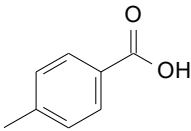
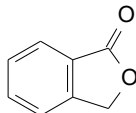
Peak ID	m/z Observed EI	MW	Tentative Assignment
A3	94,66,65,39,50	94	 or isomer
B3	105,122,77,51,50,39,38	122	 or isomer
C3	91,136,119,65,39,45,89	136	 or isomer
D3	105,77,51,50,134,38,63	134	 or isomer



FIGURE 4

A chromatogram with UV detection plus a computer reconstructed total ion chromatogram from the APCI/LC/MS/NI analysis of PTA-solid, D2403-6/3/99.

CHRO:	mt7c10109bb5	29-JUN-99	Elapse:	01:22:57	4841
Samp:	PTA slurry Solid D2403 6/3/99		Start:	14:05:09	2623
Comm:	-APCI/LC/MS 250x4.6 Zorbax Rx-C18 5u H2O/ACN/AA grad@1mL/min				
Mode:	APCI-Q3MS APICID LMR UP LR		Study:	MD-1999-010109	
Oper:	BB/TSQ700	Client: Piras	Inlet:	LC	
Peak:	1000.00 mmu	Label wndw: 1 > 2620	Masses:	120 > 1000	
Area:	0, 4.00, 0	Baseline: 0, 3	Label:	0, 40.0	

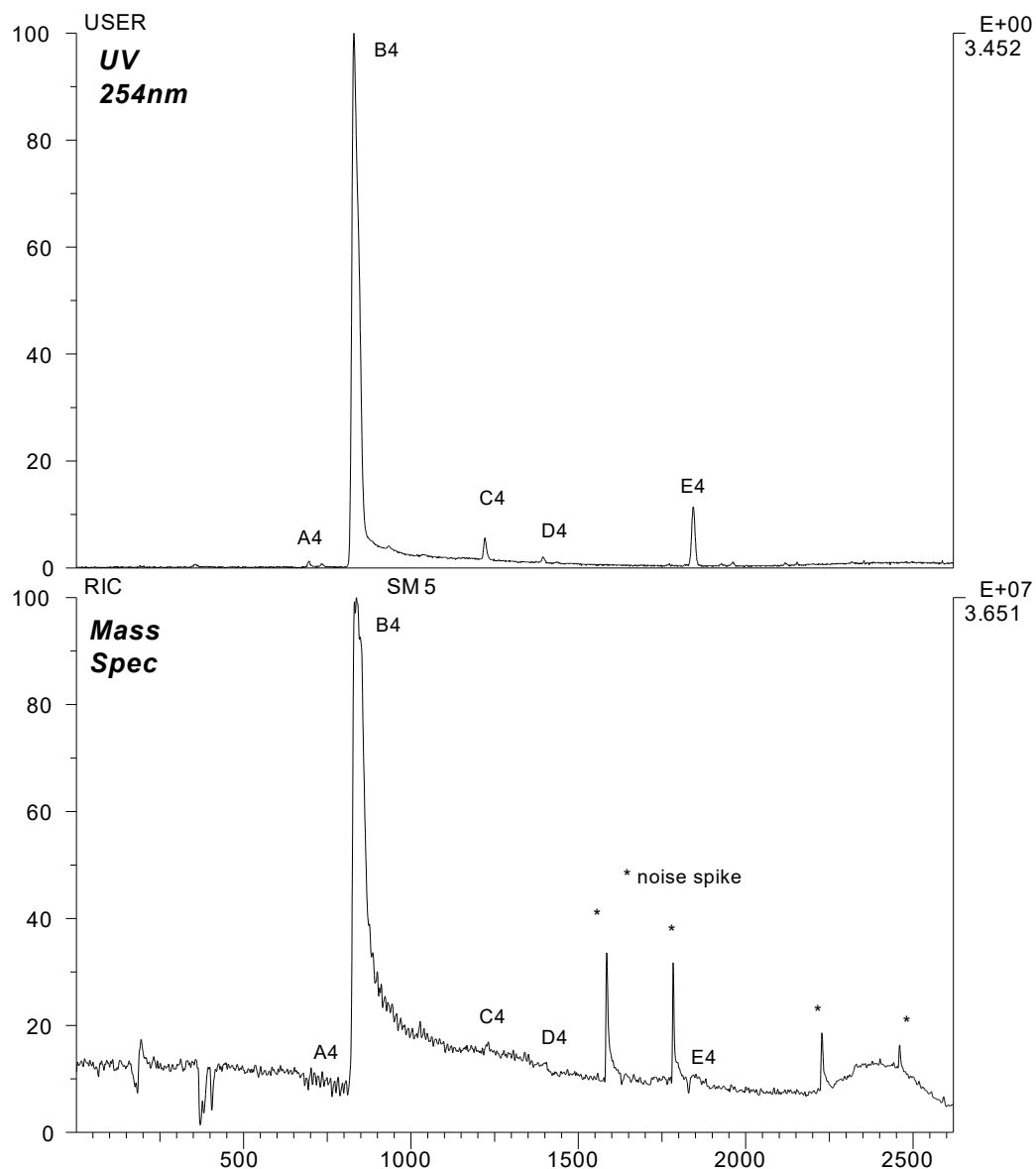


TABLE IV**Summary of mass spectral data from the APCI/LC/MS of PTA-solid, D2403-6/3/99.**

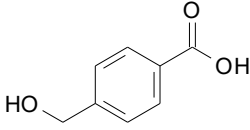
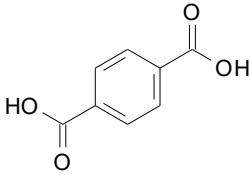
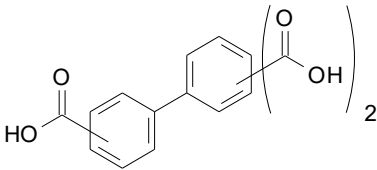
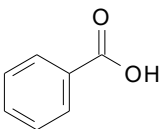
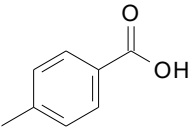
Peak ID	m/z Observed APCI/NI	MW	Tentative Assignment
A4	211,151 (M+CH ₃ CO ₂) ⁻ , (M-H) ⁻ ,	152	 or isomer
B4	165,225 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	166	 or isomer
C4	285,345 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	166	 or isomer
D4	181,121 (M+CH ₃ CO ₂) ⁻ , (M-H) ⁻ ,	122	 or isomer
E4	135,195 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	136	 or isomer



FIGURE 5

**A computer reconstructed total ion chromatogram from the GC/MS/EI analysis of
PTA-Solid, D2403-6/3/99, reacted with BSTFA.**

CHRO:	ms7k10109bb6	28-JUN-99	Elapse:	28:14	1689
Samp:	PTA-Solid D2403 (BSTFA/pyridine)		Start:	13:38:40	1533
Comm:	EI/GC/MS 30mx0.25mmx0.25u ZB-5 110-8-310 m/z 35-650@1 20/1				
Mode:	EI+Q1MS LMR UP LR		Study:	MD-1999-010109	
Oper:	BB/SSQ7000	Client: Piras	Inlet:	GC	
Peak:	1000.00 mmu	Label wndw: 1 > 1500	Masses:	35 > 650	
Area:	0, 4.00, 0	Baseline : 0, 3	Label :	0, 40.0	

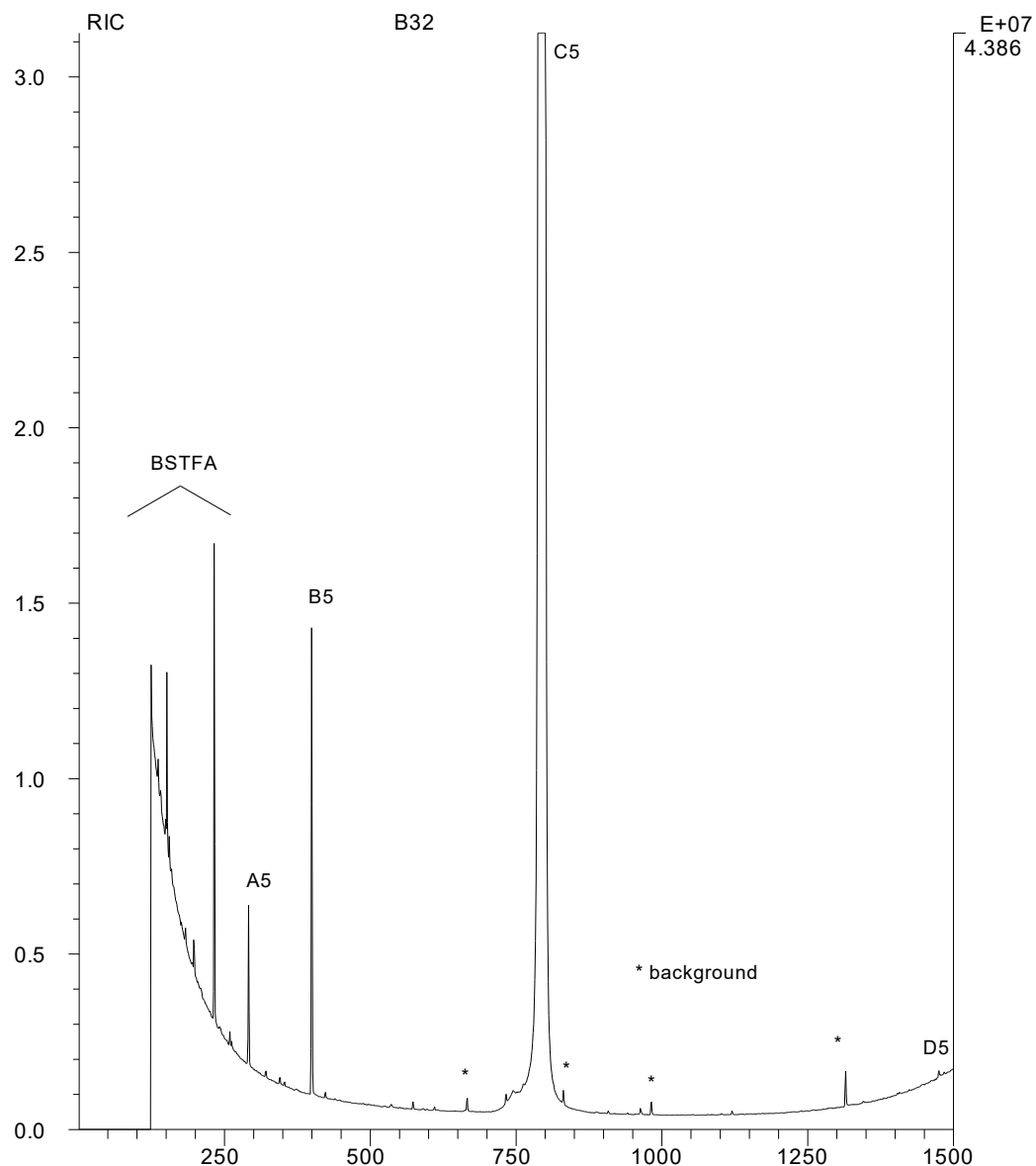


TABLE V

**Summary of mass spectral data from the GC/MS/EI and PCI-NH₃ analyses of
PTA-Solid, D2403-6/3/99, reacted with BSTFA.**

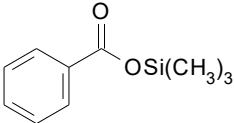
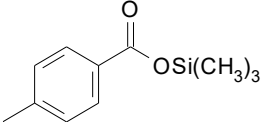
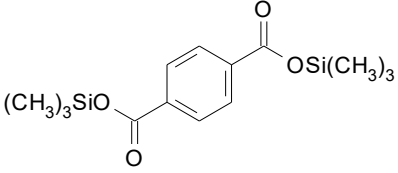
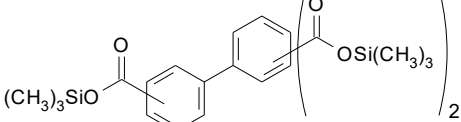
Peak ID	m/z Observed EI	m/z Observed PCI-NH ₃	MW	Tentative Assignment
A5	179,135,105,77,194,51	212,195,179,90 (M+NH ₄) ⁺ , (M+H) ⁺	194	 or isomer
B5	193,149,119,91,208,65	226,209,90,193 (M+NH ₄) ⁺ , (M+H) ⁺	208	 or isomer
C5	295,221,103,251,140,73,135,178,310	328,311,295 (M+NH ₄) ⁺ , (M+H) ⁺	310	 or isomer
D5	487,73,502,236,179,207,297,281,413,384,151	ND	502	 or isomer



FIGURE 6

A chromatogram with UV detection plus a computer reconstructed total ion chromatogram from the APCI/LC/MS/NI analysis of PTA-Condensate, E2403 6/28/99.

CHRO:	mt7c10109bb6	06-JUL-99	Elapse:	12:12.4	710
Samp:	PTA Condensate E2403 6/28/99		Start:	11:53:06	2622
Comm:	-APCI/LC/MS 250x4.6 Zorbax Rx-C18 5u H2O/ACN/AA grad@1mL/min				
Mode:	APCI-Q3MS APICID LMR UP LR		Study:	MD-1999-010109	
Oper:	BB/TSQ700	Client: Piras	Inlet:	LC	
Peak:	1000.00 mmu	Label wndw: 1 > 2620	Masses:	120 > 1000	
Area:	0, 4.00, 0	Baseline: 0, 3	Label:	0, 40.0	

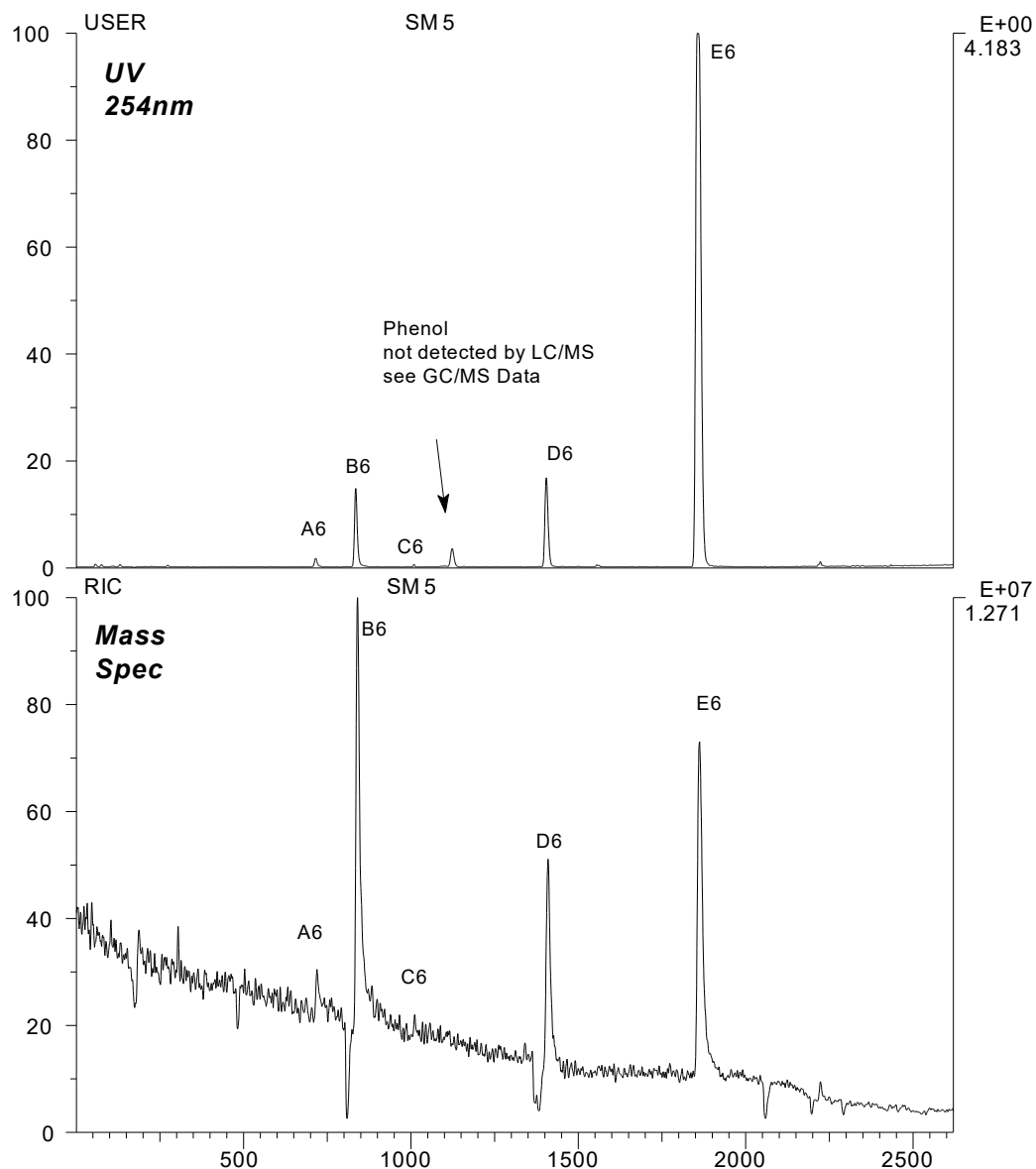


TABLE VI

**Summary of mass spectral data from the APCI/LC/MS and APCI/LC/MS/MS of
PTA-Condensate, E2403 6/28/99.**

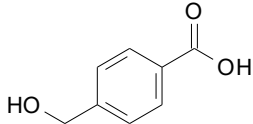
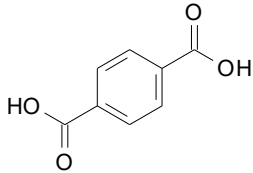
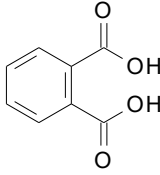
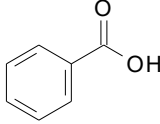
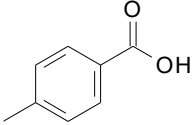
Peak ID	m/z Observed APCI/NI	MS/MS of m/z	m/z Observed daughters	MW	Tentative Assignment
A6	211,151 (M+CH ₃ CO ₂) ⁻ , (M-H) ⁻ ,	211	107,89,151, 121,77	152	 or isomer
B6	165,225 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	165	121,77,165,93	166	 or isomer
C6	165,225,331 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	165	121,165,77,93	166	 or isomer
D6	181,121 (M+CH ₃ CO ₂) ⁻ , (M-H) ⁻ ,	121	77,121,93,65	122	 or isomer
E6	135,195 (M-H) ⁻ , (M+CH ₃ CO ₂) ⁻ ,	135	91,135	136	 or isomer



FIGURE 7

**A computer reconstructed total ion chromatogram from the GC/MS/EI analysis of
PTA-Condensate, E2403 6/28/99, reacted with BSTFA.**

CHRO:	ms7k10109bb15	06-JUL-99	Elapse:	28:15	1689
Samp:	PTA-Condensate E2403 (BSTFA/pyridine)		Start:	14:10:32	2213
Comm:	EI/GC/MS 30mx0.25mmx0.25u ZB-5 110-8-310 m/z 35-650@1 20/1				
Mode:	EI+Q1MS LMR UP LR		Study:	MD-1999-010109	
Oper:	BB/SSQ7000	Client: Piras	Inlet:	GC	
Peak:	1000.00 mmu	Label wndw: 1 > 1500	Masses:	35 > 650	
Area:	0, 4.00, 0	Baseline : 0, 3	Label :	0, 40.0	

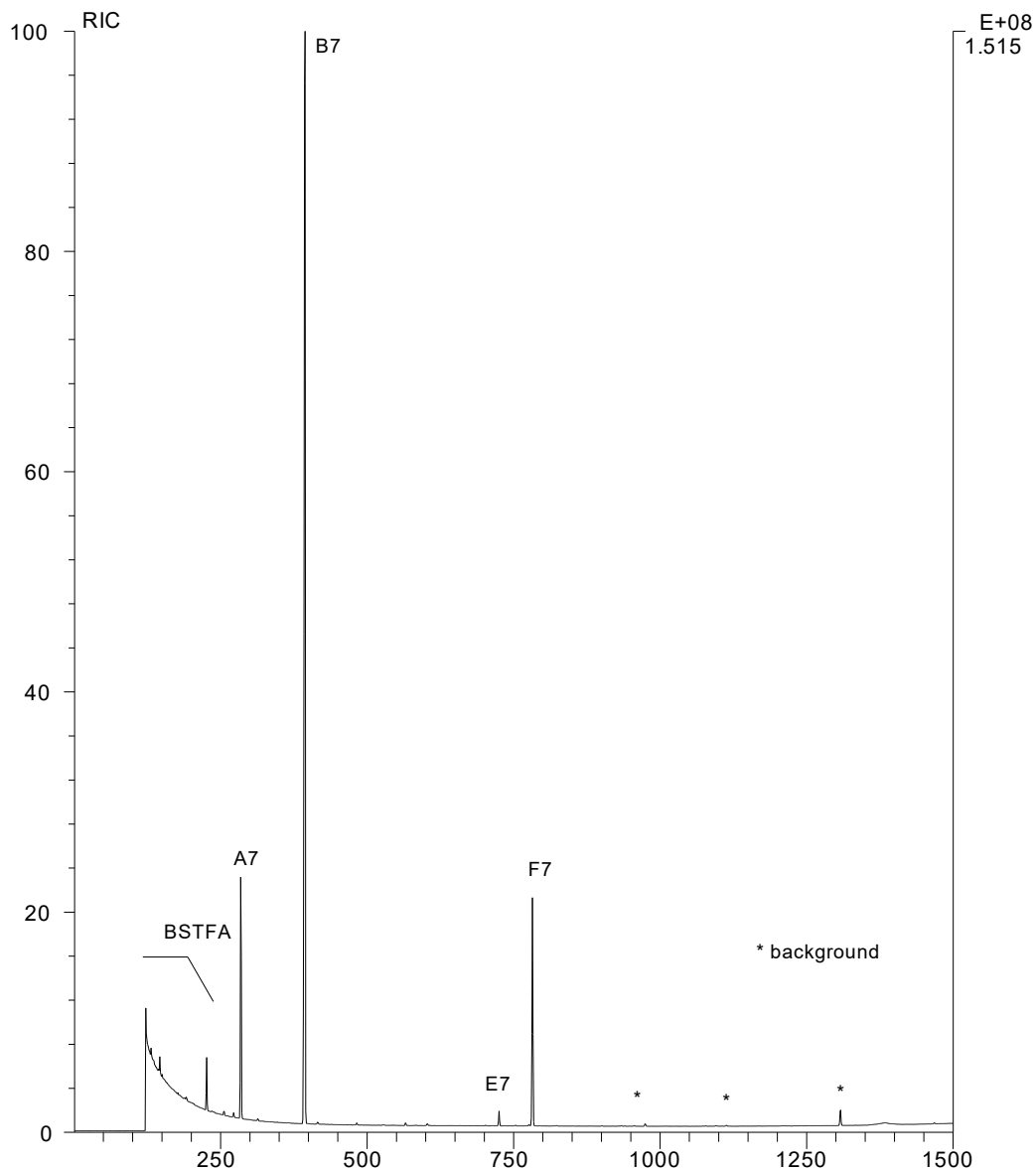


TABLE VII

**Summary of mass spectral data from the GC/MS/EI and PCI-NH₃ analyses of
PTA-Condensate, E2403 6/28/99, reacted with BSTFA.**

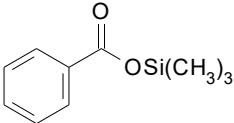
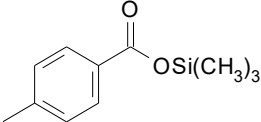
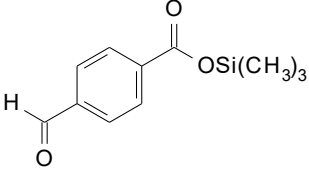
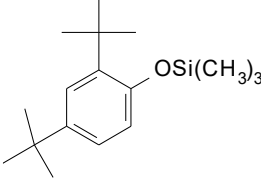
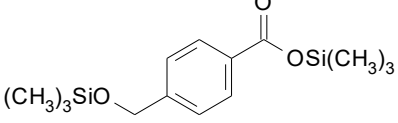
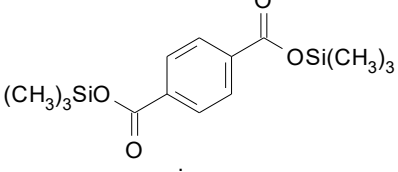
Peak ID	m/z Observed EI	m/z Observed PCI-NH ₃	MW	Tentative Assignment
A7	179,135,105,77,194,51	212,195,179,90 (M+NH ₄) ⁺ , (M+H) ⁺	194	 or isomer
B7	193,149,119,91,208,65	226,209,90,193 (M+NH ₄) ⁺ , (M+H) ⁺	208	 or isomer
C7	207,163,133,222,77,193	240,223 (M+NH ₄) ⁺ , (M+H) ⁺	222	 or isomer
D7	263,73,278,57,41	279,296 (M+H) ⁺	278	 or isomer
E7	207,281,206,73,147,179,296, 133,89,163,237,251	314,297 (M+NH ₄) ⁺ , (M+H) ⁺	296	 or isomer
F7	295,221,103,251,140,73,135, 178,310	328,311,295 (M+NH ₄) ⁺ , (M+H) ⁺	310	 or isomer



FIGURE 8

**A computer reconstructed total ion chromatogram from the GC/MS/EI analysis of
PTA-Condensate, E2403 6/28/99, extracted with methylene chloride.**

CHRO:	ms7k10109bb17	07-JUL-99	Elapse:	00:02.3	1
Samp:	PTA-Condensate E2403 (3:1 CH ₂ Cl ₂ extr)		Start:	05:50:01	992
Comm:	EI/GC/MS 30mx0.25mmx0.25u ZB-5 50-16-300 m/z 35-650@1 20/1				
Mode:	EI+Q1MS LMR UP LR		Study:	MD-1999-010109	
Oper:	BB/SSQ7000	Client: Piras	Inlet:	GC	
Peak:	1000.00 mmu	Label wndw: 1 > 1000	Masses:	35 > 650	
Area:	0, 4.00, 0	Baseline : 0, 3	Label:	0, 40.0	

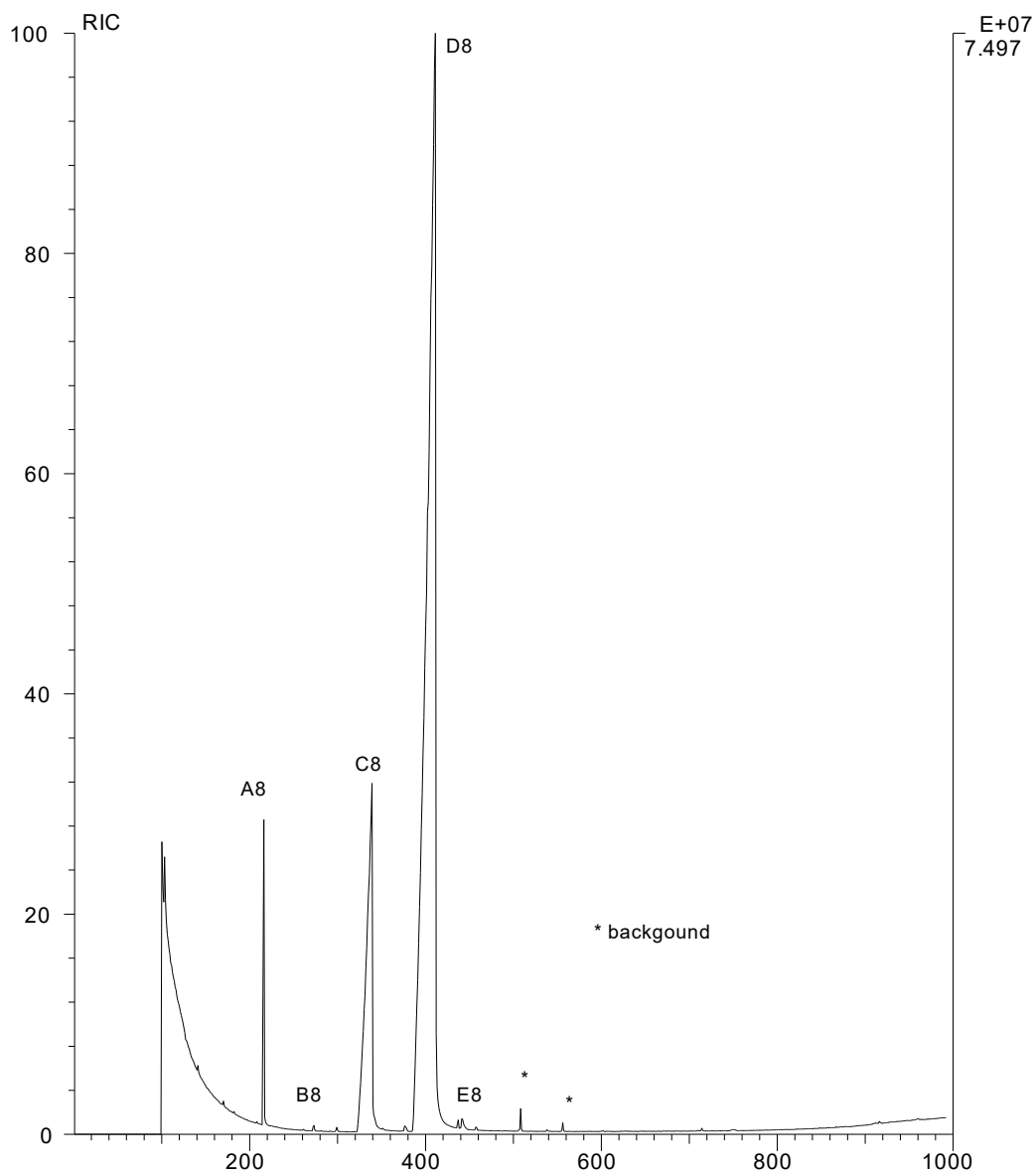
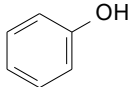
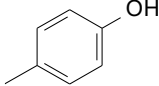
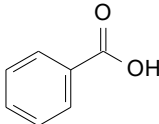
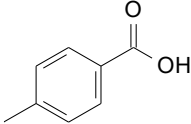
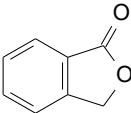


TABLE VIII

**Summary of mass spectral data from the GC/MS/EI analysis of
PTA-Condensate, E2403 6/28/99 extracted with methylene chloride.**

Peak ID	m/z Observed EI	MW	Tentative Assignment
A8	94,66,65,39,50	94	 or isomer
B8	107,108,79,77,51,39	108	 or isomer
C8	105,122,77,51,50,39,38	122	 or isomer
D8	91,136,119,65,39,45,89	136	 or isomer
E8	105,77,51,50,134,133,63	134	 or isomer

